

PLSC 309: Quantitative Political Analysis
Problem Set 1

Due: Sep-08-2026, 11:59 p.m. ET

Student Name: _____

1 Concept Checks (3 points)

Question 1 (0.3 point)

Is the following statement primarily normative or positive?

Governments should require citizens to vote in national elections.

☐ Positive

☐ Normative

Question 2 (0.3 point)

Which of the following statements is a positive statement? **(Select all that apply.)**

- A. Governments should reduce income inequality.
- B. Democracies are less likely to experience civil war than autocracies.
- C. Democracy is the best form of government.
- D. Political leaders ought to be more transparent.
- E. Voter turnout in the 2024 U.S. presidential election was approximately 64%.

Question 3 (0.3 point)

Which of the following is a clearly specified hypothesis?

- A. Social media has changed politics.
- B. Young people care about politics.
- C. Individuals who consume more political news are more likely to vote.
- D. Why do some people participate in politics more than others?

Question 4 (0.3 point)

Which of the following is NOT a clearly specified hypothesis?

- A. Higher levels of education increase political participation.
- B. Democracies are more likely to resolve disputes peacefully.
- C. Political institutions influence economic development.
- D. Individuals with higher political interest are more likely to contact elected officials.

Question 5 (0.3 point)

Consider the following hypothesis:

Individuals who trust the government are more likely to comply with public policies.

Identify the:

- Independent Variable: _____
- Dependent Variable: _____

Question 6 (0.3 point)

Consider the following hypothesis:

Countries with higher levels of trade openness are more likely to join international organizations.

Identify the:

- Independent Variable: _____
- Dependent Variable: _____

Question 7 (0.3 point)

What type of variable is the following?

Number of children in a household

- A. Nominal
- B. Ordinal
- C. Discrete
- D. Continuous

Question 8 (0.3 point)

What type of variable is the following?

Respondents rate their trust in Congress on a five-point scale:

Very low, Low, Neither low nor high, High, Very high

- A. Nominal
- B. Ordinal
- C. Discrete
- D. Continuous

Question 9 (0.3 point)

What type of variable is the following?

Political party identification:

Democrat, Republican, Independent, Other

- A. Nominal
- B. Ordinal
- C. Discrete
- D. Continuous

Question 10 (0.3 point)

What type of variable is the following?

Annual household income (measured in U.S. dollars)

- A. Nominal
- B. Ordinal
- C. Discrete
- D. Continuous

2 Understanding Political Science Research (4 points)

Read the short research article below and answer the five questions that follow. Click on the paper title to access the full text.

[Markovich, Zachary, and Ariel White. “More money, more turnout? Minimum wage increases and voting.” The Journal of Politics 84.3 \(2022\): 1834-1838.](#)

You do not need to reproduce the authors' exact wording. Write your answers in your own words based on your understanding of the paper.

2.1 Research Question (0.8 point)

- What is the main research question the paper seeks to answer?

2.2 Theory (0.8 points)

- Briefly summarize the paper's theoretical argument. Explain why the authors expect the proposed relationship to exist. (Hint: page 1834)

2.3 Hypothesis (0.8 point)

- State the paper's main hypothesis in your own words.

2.4 Conceptual Variables (0.8 points)

- What is the conceptual independent variable in the first hypothesis?
- What is the conceptual dependent variable in the first hypothesis?

2.5 Measurements (0.8 points)

The following excerpt describes one empirical analysis used in the paper. Use this excerpt to answer Question 2.5.

“We construct county-level data for presidential years from 1980 to 2016, combining data on federal and state minimum wage changes to produce a panel of the relevant minimum wage and presidential voter turnout rates for each county over this period. Then we run a variety of time series analyses to see whether increases in the minimum wage in a given place predict higher voter turnout in the next election. (p. 1837)”

- How do the authors measure the conceptual independent variable?
- Please also identify what type of variable the measured independent variable is (e.g., nominal, ordinal, discrete, or continuous).
- How do the authors measure the conceptual dependent variable?
- Please also identify what type of variable the measured dependent variable is (e.g., nominal, ordinal, discrete, or continuous).

2.6 Data Structures and Sources (Bonus, + 1 point)

Based on the research article, answer the following questions.

- (a) What is the data structure used in this study? Please explain the unit of observation. (+ 0.5 point)

- (b) What is the primary data source used in this study? (+ 0.5 point)